CHALLENGE 2 How do house prices vary across London boroughs?

Data sources : UK HPI (House Price Index), ONS (Office for National Statistics, Local Authority Districts (May 2024))

Datasets :

- UK House Price Index

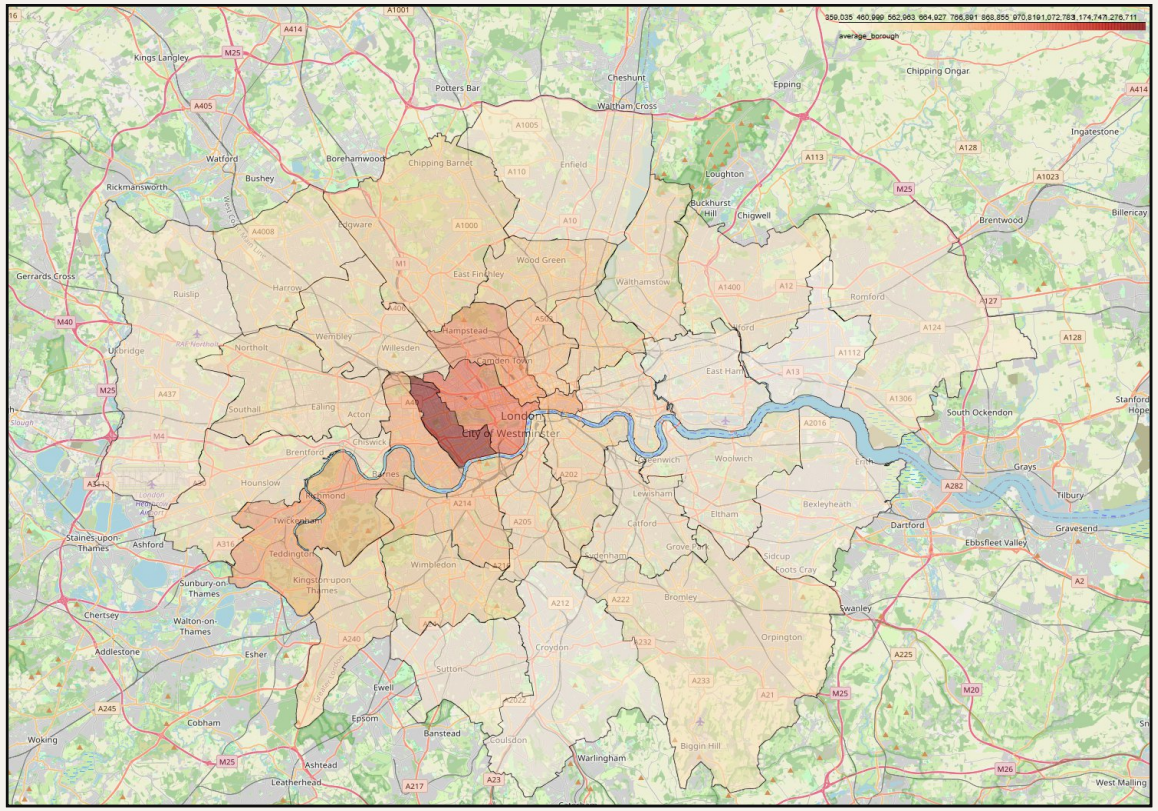
A large table containing average house prices, mortgage prices, percentage change and much more. Each row has an area code, and one entry per month from 1995 for England and Wales, 2004 for Scotland and 2005 for Northern Ireland. As this spanned over 100,000 rows, I chose to average house prices and sales volumes in each borough in 2025.

- ONS Local Authority Districts (geojson)

As my house price file did not contain any longitude or latitude data, my best option was to join the matching geometry from this shapefile. Both datasets had a column containing matching 9-digit codes unique to each Local Authority District. I renamed the columns to be identical and used pandas.merge to join on these columns, matching each average house price value to a shape.

Challenges :

- Main challenges
  - The first major difference with this data was the lack of longitude/latitude coordinates to easily plot on a map, as housing data is reported by borough. I used lessons learnt from QGIS in modules last year upon recognising the format of the 'AreaCode' column to realise that this was a seemingly straightforward join column for a LAD shapefile.
  - Oddly, the most frustrating part of this process was finding the toggle button to *apply* the filter when downloading the shapefile, as I accidentally downloaded data for the whole of the UK multiple times, and tried to pull this very large dataset from the URL, of course resulting in many request timeouts.
  - Compared to other library documentation, I found the folium documentation to be quite limited. I opted to use gdf.explore(), as this was enough for my choropleth.
- Smaller challenges/lessons
  - I learnt that matplotlib defaults to scientific notation (le6 in the corner), and I managed to solve this through : plt.ticklabel_format(style='plain', axis ='x') from here



Links : https://landregistry.data.gov.uk/app/ukhpi/
https://geoportal.statistics.gov.uk/datasets/ons::local-authority-districts-may-2024

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INSIGHTS :

A

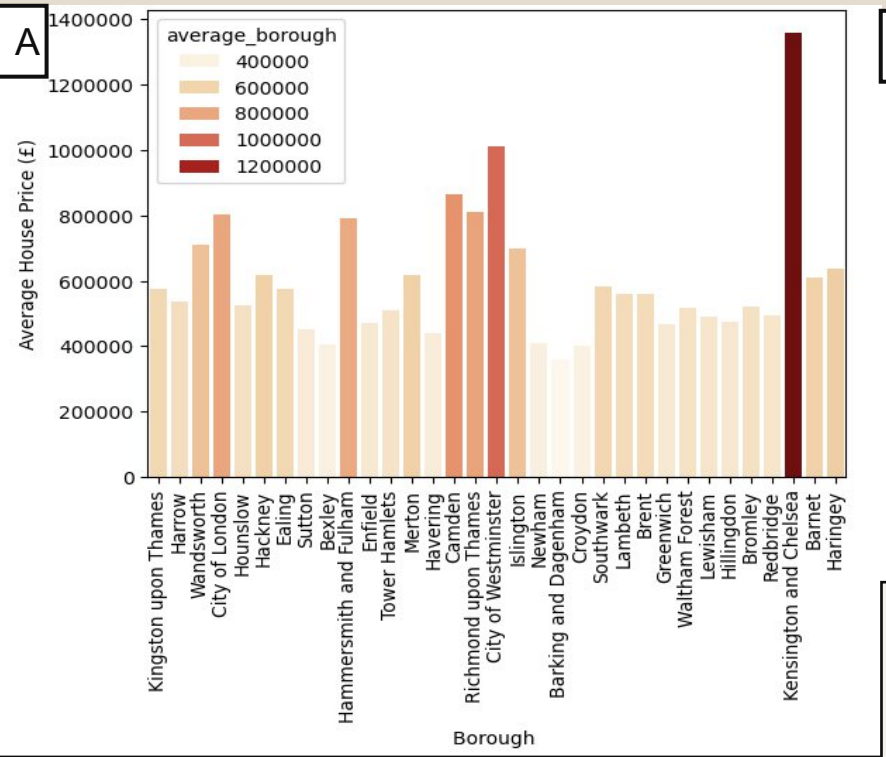


Figure A:

- The most expensive borough is Kensington and Chelsea, reporting an average house price of ~£1.3 million
- The least expensive is Barking and Dagenham at ~£350,000.
- When comparing this with my map, it is clear that central boroughs (Camden, City of Westminster, Hammersmith and City of London) are significantly more expensive on average than outer areas.

B



Figure B:

- Two key outliers of ~£1 million and ~£1.3 million, in City of Westminster and Kensington and Chelsea respectively.
- These trends in central London may be a combination of London's global presence, attracting particularly wealthy individuals, Royal heritage (proximity to Kensington Palace), and land scarcity making new developments especially rare.
- In surrounding boroughs, though, these prices tend to range between ~£500,000-600,000 with the median being ~£550,000.
- It is important to note that this box plot reports the average price in each borough for all of 2025, each point reflects one borough's average price.

C

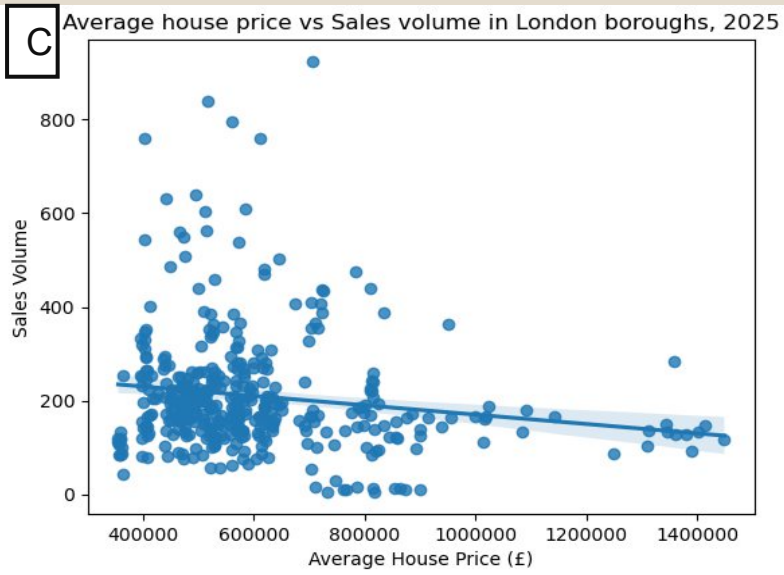


Figure C:

- Unlike Figures A and B, this scatter uses all points from each borough in each month.
- There is a weak negative relationship between sales volume and house prices, with higher priced areas having less sales.
- Several outliers are present, in both sales volume and average price.

D

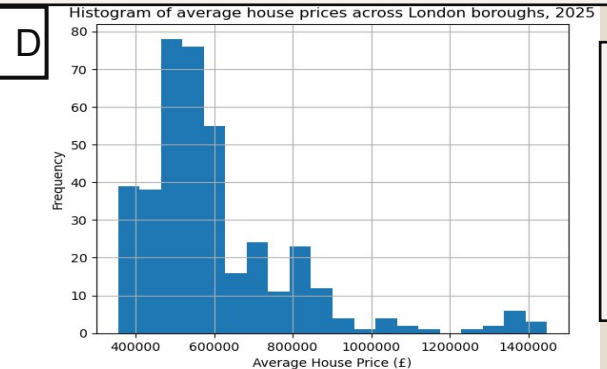
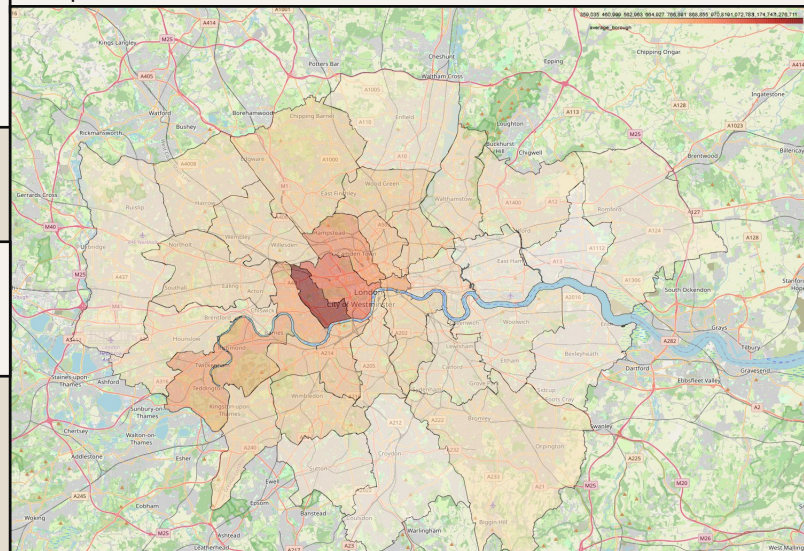


Figure D

- Histogram displays all months per borough across 2025.
- Prices are skewed to the right, most boroughs have an average price of ~£400-000-600,000.
- The upper tail shows a cluster of average prices around ~£1.4 million, which we can assume to be in boroughs coloured darker reds on both the choropleth map and bar chart,
- This tail explains the mean being higher than the median in descriptive statistics

How do house prices vary across London boroughs?



CHALLENGE 4

Which areas of the Americas have the highest concentration of satellite-detected thermal hotspots?

Data source : NASA FIRMS Satellite Active Fire data : https://www.earthdata.nasa.gov/data/tools/firms

Dataset:

This dataset was taken from NASA's FIRMS (Fire Information for Resource Management System) and is a large collection of point data taken from the MODIS and VIIRS satellites, listing detections of active fires and thermal anomalies across the globe.

The original dataset contains over 5m+ rows, logging the intensity, longitude, latitude, time, FRP (fire radioactive power) and assumed hot spot type (volcano, offshore, land source etc.) of each thermal anomaly since November 2000 at 1km (MODIS) or 375m (VIIRS) resolution.

Challenges :

- Large dataset

This dataset was very large in size, and downloads are limited. To download the data, an account must be created and a request must be submitted to retrieve a specific portion of the dataset. These limits meant that I had to subset my region to North and South America, and only take records from 2004-2026, as this was within the download request limit.

As per the lab instructions, I used datashader to explore and plot this dataset. The original download size was ~1.6GB and in .csv format. To avoid hosting this file through a cloud service, I decided to take a sample of this data and save this as a separate parquet file, to be uploaded and used when running the notebook.

- Projection mismatch

I repeatedly ran into an error where Esri tiles were grey, filled with 'map data not yet available'. After researching this error and comparing my data to the nyc_taxi data in the lab, I remembered that Esri, OSM and other online map services often use Web Mercator projection rather than EPSG:4326. I had to install geoviews and update libraries in order to get a map output.

- RAM errors / timeouts

My kernel repeatedly crashed or timed out throughout this challenge, and my task manager reported 98% RAM usage which I assume to be the culprit. To get around this, I was able to copy my code and required libraries into a separate notebook to avoid loading any unnecessary cells, and I sampled the data to 100,000 rows.

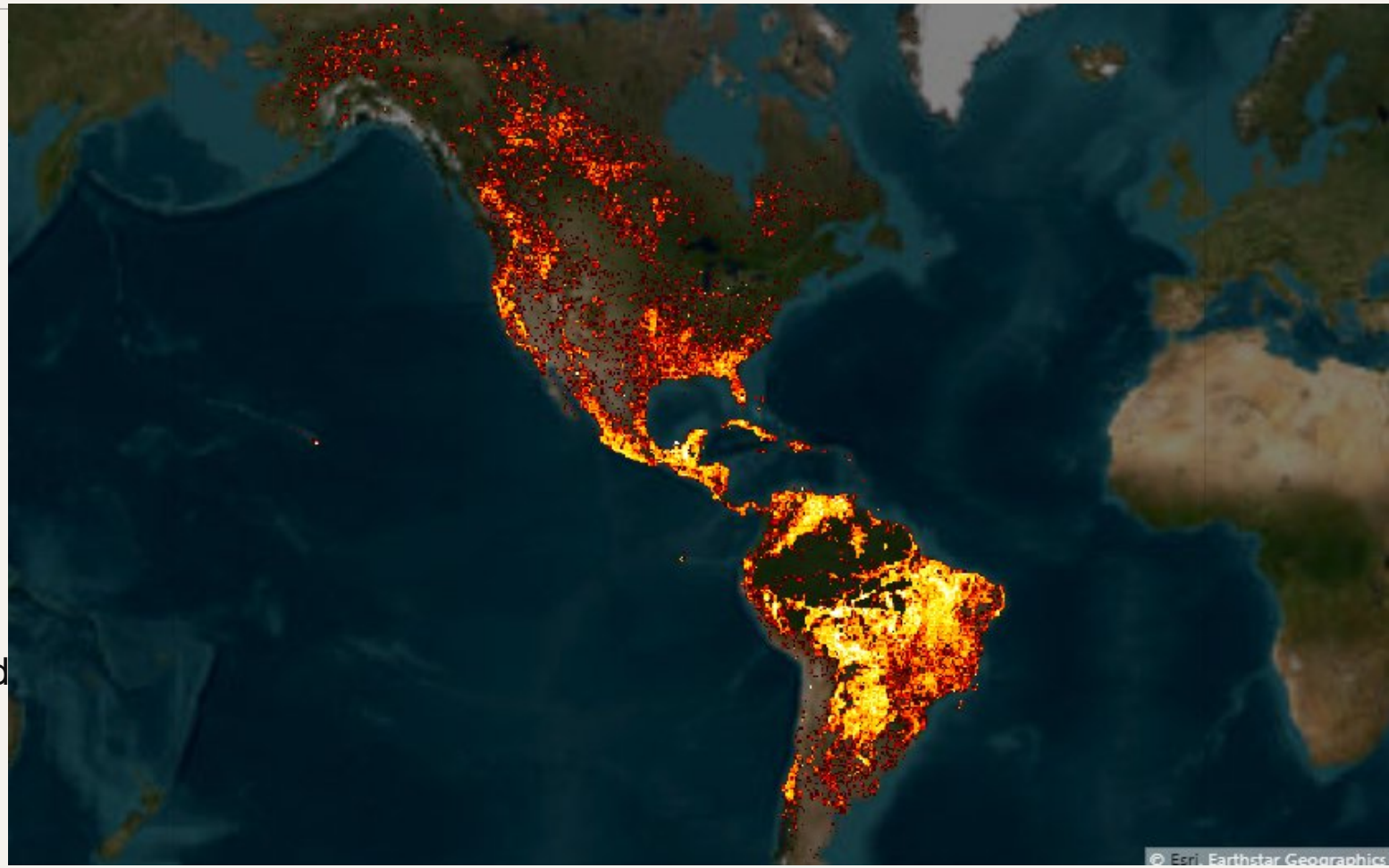
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Problem / Scenario

Using this dataset, you could display different columns to explore the spatial distribution of fire hotspots over time, whether fires tend to be more powerful or hotter (in brightness or frp columns) in different areas, or whether they occur more in day or night.

If you were to focus in on a specific city or state, you could also use this data in combination with land-use data, and measure whether instances of fires are more likely on certain land-cover types (e.g dry/barren land as opposed to farmland).

Satellite Detections of Thermal Anomalies in the Americas, 2004-2026



Findings

From the map, the brightest pixels are concentrated in areas of central and eastern Brazil, and in surrounding areas of the Amazon rainforest. There are also very bright pixels across the Gulf of Mexico and across Mexico.

There are some thermal hotspots scattered across Canada and Alaska, but fires tend to concentrate in the South/South-East of North America and are more scattered across the North.

It is important to note that this map is a 100,000 point sample taken from the original dataset, and may not accurately represent spatial trends in the region, as a very large chunk of data has been removed. This dataset may also include instances of intentional fires (e.g crop burning), as these are still considered thermal anomalies when detected by the satellite.

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